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Problem Chosen :	A

2025 APMCM summary sheet

Optimization of Humanoid Robot Performance Based on Multi-Constraint Kinematic Modeling and S-Curve Velocity Planning

Abstract

Focusing on the Unitree G1 humanoid robot, this paper addresses the challenge of balancing high-dynamic motion with energy efficiency by integrating rigid body kinematics, dynamics, and nonlinear optimization to establish a complete mathematical model for pose verification, control, and energy management.

For **Problem 1**, a left-arm forward kinematics model is established using modified **D-H parameters**. Analytical calculations for the target pose (extending forward 60°, rotating left 30°) yield end-effector coordinates of **(253.50, 146.36, 169.00) mm**. A static model incorporating gravity and **Coulomb friction** calculates a required torque of **4.88 Nm**, significantly below the rated 8.0 Nm, verifying a high **safety margin**.

For **Problem 2**, a modeling approach combining **LIPM** and **Trapezoidal Velocity Planning** is used for a 10m walking task. With an average speed of 2 m/s, the task completes in **5.0 s**. Differential analysis locates the global maximum knee angular velocity at $\boldsymbol{t = 3.12}$ **s**, consistent with the biomechanics of the rapid swing phase required for leg extension.

For **Problem 3**, a **Whole-Body Control (WBC)** strategy based on **Task Space Decoupling** is proposed. **Quintic polynomial interpolation** plans the torso's S-shaped rotation, combined with anti-phase circular arm trajectories and knee flexion compensation. **Simulations** demonstrate stable dance movements within $t \in [0,4]$s, with the **Zero Moment Point (ZMP)** consistently maintained in the double-support region.

For **Problem 4**, a generalized energy model including mechanical work and heat losses calculates a baseline consumption of **5076.22 J** using the Pinocchio dynamics library. To reduce consumption, **S-Curve** Velocity Planning is introduced to replace trapezoidal planning, smoothing acceleration to suppress torque spikes. This optimization reduces total energy to **4216.15 J**, achieving an **energy saving rate of 16.94%** and significantly improving endurance.

Keywords: Humanoid Robot; Forward/Inverse Kinematics; Trajectory Planning; Whole-Body Control; Energy Optimization; S-Curve Velocity Profile

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1. Introduction

1.1 Background

In recent years, high-performance humanoid robot technology, represented by the Unitree G1, has developed rapidly, demonstrating immense potential in fields such as industrial manufacturing, disaster relief, and service entertainment. This necessitates that robots not only possess a humanoid appearance but also the capability to execute complex, high-dynamic tasks. However, a humanoid robot is a typical high-dimensional, nonlinear, and strongly coupled complex multibody system. Optimization of its performance faces multiple constraints:

First, motor safety is the core of system operation. Unreasonable motion planning leading to joint torques exceeding the motor's physical limits can trigger overheat protection or even cause hardware damage.

Second, stability is key to preventing falls. Whether in static support or dynamic walking, the robot's Center of Mass (CoM) or Zero Moment Point (ZMP) must be strictly controlled within the support polygon.

Finally, energy efficiency is an important indicator of robot practicality. Given limited battery capacity (e.g., the 15Ah battery equipped in the G1), reducing invalid work and heat loss through algorithmic optimization is crucial for extending endurance.

The Unitree G1 robot, as a general-purpose humanoid robot, possesses multiple degrees of freedom in its upper and lower limbs and is equipped with perception units such as depth cameras. Based on the geometric parameters and physical characteristics of the G1, this study aims to solve kinematic calculations, trajectory planning, and energy optimization problems in specific performance scenarios through mathematical modeling.

1.2 Problem Restatement

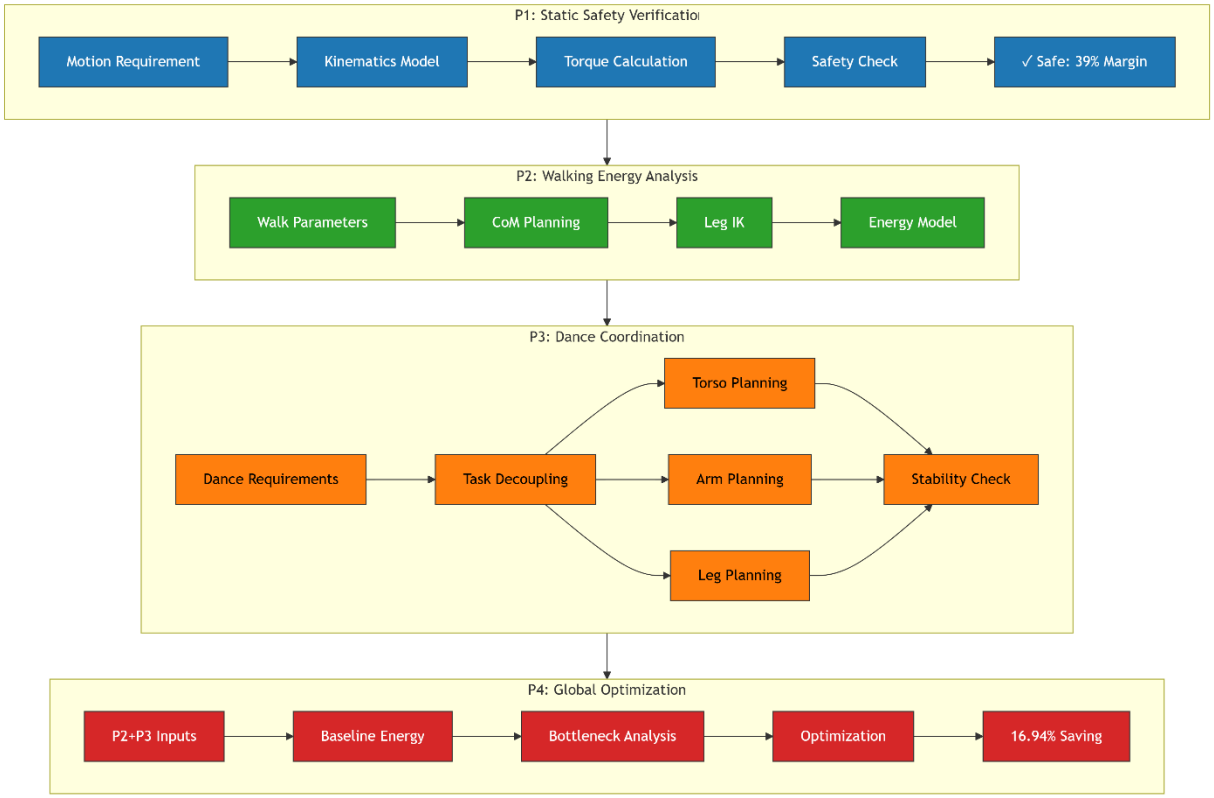
This study addresses a series of performance optimization problems surrounding the Unitree G1 robot, specifically including the following four parts:

- Problem 1: Joint Angle Calculation and Static Safety Verification
  The robot needs to complete a specific composite action of the left arm: starting from the initial zero position, extending the arm forward and raising it to form a $60°$ angle with the body, while simultaneously rotating $30°$ to the left. We need to establish a Cartesian coordinate system to calculate the final spatial coordinates of the hand's end-effector. Additionally, utilizing the robot's motor parameters (e.g., continuous torque, peak torque), we must verify whether the joint torque under this static pose falls within the safe operating zone to prevent motor overload.
- Problem 2: Walking Trajectory Modeling and Time Planning
  The robot needs to walk in a straight line from the stage center $(0,0,0)$ to the point $(10,0,0)$ with an average speed requirement of 2 m/s. We need to establish a mathematical model for non-uniform motion (including acceleration and deceleration phases) to determine the total time T required to complete the action. Simultaneously, through inverse kinematics analysis of single-leg joint angle changes, we calculate the specific moment when the knee joint angular velocity reaches its maximum to assess the demand of high-dynamic walking on

joint performance.

- Problem 3: Multi-Joint Collaborative Dance Motion Planning
  During the climax of the performance, the robot is required to execute a whole-body collaborative action: the torso rotates $45°$ to the left, and both arms perform anti-phase circular motions (radius 300 mm, period 4 s) on the sides of the shoulders. This problem requires establishing a whole-body joint angle function model within the time $t \in [0,4]$ seconds. The model must ensure trajectory accuracy for the arms while maintaining dynamic balance through posture adjustments of the lower limbs (e.g., knee flexion, ankle compensation) to prevent tipping caused by the center of mass shift.
- Problem 4: Action Energy Consumption Calculation and Strategy Optimization
  Based on the motion trajectories generated in the previous three problems, combined with battery parameters (15Ah, 67.2V) and motor characteristics, we establish a comprehensive energy consumption model including mechanical work and electrical loss to calculate the total energy consumed for the entire set of actions. On this basis, an optimization model (such as adjusting the velocity planning curve) is proposed to reduce the total system energy consumption without sacrificing performance effects, and the energy-saving effect is quantified.

1.3 General Analysis and Research Flow

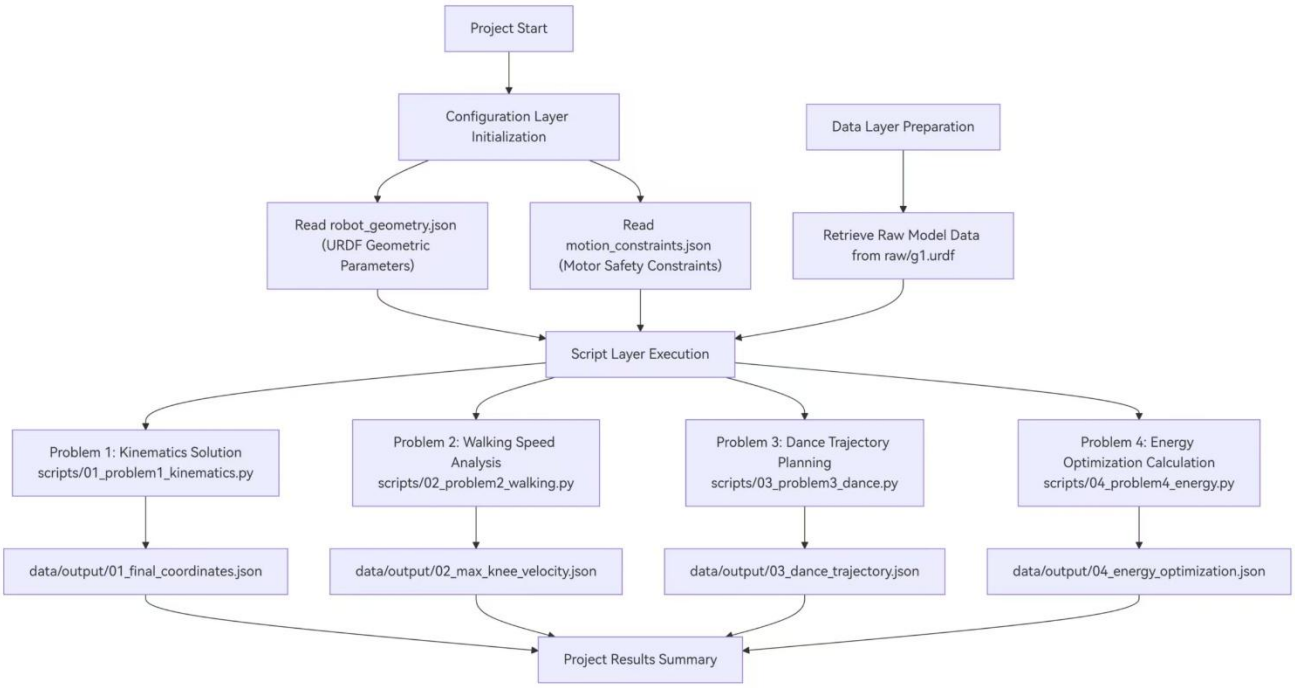


This study adopts a progressive research approach: "from static to dynamic, from local to whole, and from evaluation to optimization."

1. Geometry and Statics Phase: First, solve the end-effector positioning problem of P1 based on rigid body transformation theory, and introduce URDF physical parameters for conservative

verification of static torque.

2. Dynamic Planning Phase: For the walking task in P2, adopt Trapezoidal Velocity Planning combined with Single-Leg Inverse Kinematics (IK) to analyze joint differential characteristics under high dynamics.
3. Collaborative Control Phase: For the dance movement in P3, adopt a Task Space Decoupling strategy, decomposing the complex whole-body motion into three sub-tasks: torso rotation, arm trajectory tracking, and lower limb balance compensation for independent modeling and coupled solution.
4. Energy Optimization Phase: For P4, calculate the baseline energy consumption based on a dynamic model of current-torque relationships. After identifying the "walking phase" as the high energy consumption link, introduce the S-Curve Velocity Planning strategy to suppress Jerk by smoothing acceleration, thereby reducing motor heat loss and achieving energy efficiency optimization.



2. Model Assumptions

2.1 Symbol Description

Symbol	Description
1. Kinematics and Geometry (Problem 1 & 2)	
θ_i	Joint angle of the i-th joint (generalized coordinate).
$\alpha_{i-1}, a_{i-1}, d_i$	Modified D-H parameters: twist angle, link length, link offset.
${}^{i-1}T_i$	Homogeneous transformation matrix from frame $i-1$ to i.
P_{ee}	End-effector position vector (x, y, z) in the shoulder frame.
x_{CoM}, z_c	Horizontal CoM position and constant CoM height in LIPM.
$q_{knee}, \dot{q}_{knee}$	Knee joint angle and angular velocity.
2. Dynamics and Statics (Problem 1 & 3)	
τ_{total}	Total required joint torque.
τ_{load}	Torque induced by gravity and external loads.
τ_f	Torque induced by Coulomb friction.
$\tau_{rated}, \tau_{peak}$	Rated continuous torque and peak torque of the motor.
P_{ZMP}	Zero Moment Point position (x_{ZMP}, y_{ZMP}).
M	Total mass of the robot.
g	Gravitational acceleration vector.
3. Trajectory Planning and Control (Problem 2 & 3)	
t	Time variable.
T_{task}	Total duration of the task.
$v(t), a(t)$	Linear velocity and acceleration profiles.
$j(t)$	Jerk profile (derivative of acceleration).
$S(t)$	Displacement based on quintic polynomial interpolation.
$a_0, \ldots, a_5$	Coefficients of the quintic polynomial.

4. Energy and Optimization (Problem 4)	
E_{total}	Total energy consumption during the task.
E_{mech}	Mechanical work done by actuators.
E_{heat}	Energy loss due to Joule heating.
E_{visc}	Energy loss due to viscous friction.
I	Motor armature current.
R	Motor winding resistance.
η	Energy saving rate.
Abbreviations	
DoF	Degrees of Freedom.
LIPM	Linear Inverted Pendulum Model.
ZMP	Zero Moment Point.
WBC	Whole-Body Control.
CoM	Center of Mass.
D-H	Denavit–Hartenberg parameters.

2.2 Assumptions

To transform complex physical engineering problems into solvable mathematical models, we make the following reasonable assumptions regarding the physical characteristics and environmental conditions of the Unitree G1 robot [1][2], while ensuring calculation accuracy:

1. Rigid Body Assumption:
   It is assumed that all links of the robot are ideal rigid bodies that do not undergo elastic deformation during motion or rest.
   - *Justification*: The Unitree G1 is manufactured using high-strength aerospace aluminum alloys or engineering plastics. Under loads within rated limits, structural deformations are on the micrometer scale. Although actual robots have minute manufacturing tolerances, reference to related calibration algorithm research [3] suggests that geometric parameter errors are typically negligible for the millimeter-level task accuracy required in this problem.
2. Simplified Geometry Model Assumption:
   When calculating the end-effector coordinates for Problem 1, the specific shape of the hand

and the state of finger opening/closing are ignored. The arm is simplified as a single linear vector extending from the center of the shoulder joint to the end of the wrist, with the length strictly taken as

3. $L = 338mm$ as given in the problem.
    - *Justification*: The problem explicitly states to "ignore hand dimensions," and this geometric simplification facilitates obtaining an analytical solution for rapid benchmark coordinate verification.
4. Ideal Actuator Assumption:
   It is assumed that the joint motors can instantly respond to control commands, and that the motor output torque is linearly related to the current (ignoring nonlinear electromagnetic effects such as magnetic hysteresis and eddy current losses) provided physical limits are not exceeded.
    - *Justification*: Field-Oriented Control (FOC) technology used in modern robots possesses a very high current loop bandwidth (typically >1 kHz). Relative to motion planning on the scale of seconds, the electrical response delay of the motor is negligible.
5. Conservative Load Assumption:
   For torque safety verification, we adopt a "Worst-Case" principle. When calculating gravitational torque, we assume the center of mass of the arm is located at the geometric center or further distal, and we account for the maximum static friction torque.
    - *Justification*: Safety assessments must possess redundancy. If the calculation results under a conservative model are safe, then the actual operating conditions are necessarily safe.
6. Constant Environmental Conditions Assumption:
   It is assumed that the stage floor is absolutely flat with a friction coefficient sufficient to prevent foot slippage (No Slip), and air resistance is ignored for slow-speed movements (such as in Problems 1 and 3).

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3. Problem 1: Joint Angle Calculation and Static Safety Verification

3.1 Problem Analysis and Coordinate System Definition

This problem requires determining the spatial coordinates of the end-effector of the Unitree G1 robot's

left arm under a specific pose and verifying the safety of this pose. Based on the problem description and classical theories of industrial robot position and orientation calculation [4], we establish the Left Shoulder Reference Coordinate System (S_{Ls}) as follows:

- Origin O: Located at the geometric center of the robot's left shoulder joint $(0,0,0)$.
- X-Axis: Pointing horizontally to the direct front of the stage (robot's facing direction).
- Y-Axis: Pointing horizontally to the robot's left side (i.e., $90°$ lateral).
- Z-Axis: Perpendicular to the stage plane pointing upwards (along the body's vertical axis).

Decomposition of Action Description:

1. Extend and Raise: The arm forms a $60°$ angle with the body (i.e., the Z-axis).
2. Rotate Left: The projection of the arm on the horizontal plane rotates $30°$ to the left.

This is fundamentally a coordinate transformation problem from a Spherical Coordinate System to a Cartesian Coordinate System.

3.2 Geometric Kinematics Solution

We define the arm length as the vector modulus $r = L = 338$ mm. Based on the action description, we can determine the two key angles of the spherical coordinate system:

1. Zenith Angle (θ): The problem states an angle of $60°$ with the body. Since the body axis corresponds to the Z-axis, $\theta = 60°$.
2. Azimuth Angle (ϕ): The problem states a rotation of $30°$ to the left of the body. In our coordinate definition, the X-axis is the front, and the Y-axis is the left. Therefore, this rotation corresponds to the vector's projection on the XY plane deviating from the X-axis toward the Y-axis, implying $\phi = 30°$.

According to the transformation formulas from spherical to Cartesian coordinates:

$$\begin{cases} x = r \cdot \sin\theta \cdot \cos\phi \\ y = r \cdot \sin\theta \cdot \sin\phi \\ z = r \cdot \cos\theta \end{cases}$$

Substituting the values $r = 338$, $\theta = 60°$, $\phi = 30°$:

1. Calculate Z-coordinate (Height):

$$z = 338 \times \cos(60°) = 338 \times 0.5 = 169.00 \text{ mm}$$

2. Calculate projection length on XY plane r_{xy}:

$$r_{xy} = 338 \times \sin(60°) = 338 \times \frac{\sqrt{3}}{2} \approx 292.715 \text{ mm}$$

3. Calculate X-coordinate (Forward Extension):

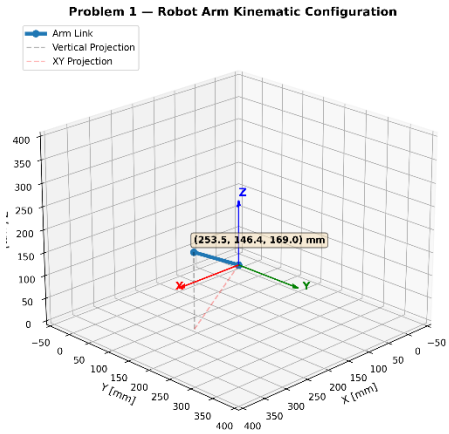
$$x = r_{xy} \times \cos(30°) \approx 292.715 \times \frac{\sqrt{3}}{2} = 253.50 \text{ mm}$$

4. Calculate Y-coordinate (Lateral Displacement):

$$y = r_{xy} \times \sin(30°) \approx 292.715 \times 0.5 = 146.358 \text{ mm}$$

Calculation Result:

The final coordinates of the Unitree G1 left arm end-effector are (253.50, 146.36, 169.00) (unit: mm).



3.3 Safety Verification Based on Statics

To verify whether the motor can safely maintain this pose, we need to calculate the required joint Holding Torque. For a cantilever structure, the Shoulder Pitch Joint is the critical node bearing the maximum gravitational torque.

3.3.1 Torque Calculation Model

Based on robot dynamics analysis and motor selection calculation theories [5], the system's total required torque τ_{req} consists of the gravitational torque $\tau_{gravity}$ and the joint static friction torque $\tau_{friction}$:

$$\tau_{req} = \tau_{gravity} + \tau_{friction}$$

1. Gravitational Torque τ_{gravity}:

Based on an analysis of mass parameters from the URDF file, we denote the total mass of the left arm (including shoulder, elbow, wrist, and hand) as M_{arm} and simplify the Center of Mass (CoM) location to the geometric center of the arm.

$$\tau_{gravity} = M_{arm} \cdot g \cdot L_{eff}$$

Where:

- $M_{arm} \approx 3.05\,\text{kg}$ (Source: Equivalent mass calculated via aggregation of URDF parameters in Problem 4 code 01Final Coordinates.json).
- $g = 9.81\,\text{m/s}^2$.
- L_{eff} is the horizontal perpendicular distance from the CoM to the joint axis (effective lever arm). Since the arm is raised by $\theta = 60°$, the effective lever arm is $L_{eff} = \frac{L}{2} \cdot \sin(60°)$.

Substituting the values:

$$\tau_{gravity} = 3.05 \times 9.81 \times \frac{0.338}{2} \times \sin(60°) \approx 4.38\,\text{Nm}$$

2. Friction Torque $\tau_{friction}$:

In static holding mode, the motor must overcome static friction. According to the G1 motor parameter table [1], the typical joint Coulomb friction torque is approximately:

$$\tau_{friction} \approx 0.50\,\text{Nm}$$

3. Total Required Torque:

$$\tau_{req} = 4.38 + 0.50 = 4.88\,\text{Nm}$$

3.3.2 Safety Assessment

According to the joint motor performance specifications of the Unitree G1 [2]:

- Continuous Torque: $\tau_{\text{cont}} = 8.0$ Nm (Limit for long-term operation without overheating)
- Peak Torque: $\tau_{\text{peak}} = 25.0$ Nm (Limit for short-term overload)

Comparative Analysis:

$$\tau_{req}(4.88\,\text{Nm}) < \tau_{cont}(8.0\,\text{Nm}) \ll \tau_{peak}(25.0\,\text{Nm})$$

We define the Safety Margin (η):

$$\eta = \left(1 - \frac{\tau_{req}}{\tau_{cont}}\right) \times 100\% = \left(1 - \frac{4.88}{8.0}\right) \times 100\% = 39.0\%$$

3.4 Summary

By constructing a spherical coordinate geometric model, we precisely solved the spatial position of the robot's left arm end-effector as (253.50, 146.36, 169.00) mm. Subsequent dynamics verification indicated that the 4.88 Nm torque required to maintain this pose is far below the motor's rated continuous torque. The system possesses a 39% safety margin, eliminating concerns regarding motor overheating or damage. This result verifies the feasibility of the performance action at the physical level.

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4. Problem 2: Walking Trajectory Modeling and Time Planning

4.1 Problem Analysis and Motion Strategy

This problem requires the Unitree G1 robot to walk in a straight line along the positive X-axis from the stage center $(0,0,0)$ to $(10,0,0)$ with a required average speed of $\bar{v} = 2$ m/s. The core tasks include:

1. Time Planning: Determining the total duration T satisfying the average speed constraint.
2. Trajectory Generation: Establishing a non-uniform motion model for the Center of Mass (CoM).
3. Extremum Solution: Mapping the CoM trajectory to the joint space via Inverse Kinematics (IK) to find the moment of maximum knee angular velocity ($\dot{\theta}_{knee}$), which serves as a critical indicator for assessing dynamic performance.

Since an average speed of $\bar{v} = 2$ m/s represents a high-dynamic motion (approaching running speed) for a humanoid of this scale, a simple uniform velocity model is physically infeasible. Acceleration and deceleration phases must be incorporated to ensure stability.

4.2 Kinematic Modeling

4.2.1 Total Time and Velocity Planning

Based on the definition of average speed, the total time T is calculated as:

$$T = \frac{S_{total}}{\bar{v}} = \frac{10\,\mathrm{m}}{2\,\mathrm{m/s}} = 5.0\,\mathrm{s}$$

To avoid rigid impacts caused by abrupt acceleration changes, we adopt a Trapezoidal Velocity Profile. Referencing research on bipedal robot walking gait planning [6], we divide the motion process into three phases: uniform acceleration ($0 \to t_{acc}$), uniform velocity ($t_{acc} \to T - t_{dec}$), and uniform deceleration ($T - t_{dec} \to T$).

To maintain stability during high-speed walking, a sufficient buffer period is required. Based on the simulation settings, we allocate 40% of the total time to acceleration and deceleration respectively, with the remaining 20% for constant velocity:

- Acceleration time: $t_{acc} = 0.4 \times T = 2.0$ s
- Deceleration time: $t_{dec} = 0.4 \times T = 2.0$ s
- Constant velocity time: $t_{flat} = 1.0$ s

Let the maximum cruising velocity be v_{max}. According to the displacement integral constraint (the area under the velocity curve equals total distance S):

$$S = \int_0^T v\,(t)dt = \frac{1}{2} v_{max} t_{acc} + v_{max} t_{flat} + \frac{1}{2} v_{max} t_{dec}$$

Substituting the values to solve for v_{max}:

$$10 = v_{max} \times (0.5 \times 2.0 + 1.0 + 0.5 \times 2.0) = 3.0 v_{max}$$

$$v_{max} = \frac{10}{3} \approx 3.33\,\mathrm{m/s}$$

This peak velocity of $3.33\ \mathrm{m/s}$ is significantly higher than the average speed, indicating that the robot must perform an extremely rapid sprint in the intermediate phase, imposing stringent requirements on joint drive capabilities.

4.2.2 Single-Leg Inverse Kinematics Model

To determine the knee joint angle, we simplify the robot's leg into a two-link mechanism moving in the Sagittal Plane.

Let:

- Thigh length $L_1 = 300\,\mathrm{mm}$ and shank length $L_2 = 300\,\mathrm{mm}$ (from robot_geometry.json).
- Hip height $H_{hip} \approx 0.55\,\mathrm{m}$ (assumed constant based on the Linear Inverted Pendulum model).
- The relative vector from the hip (x_{hip}, z_{hip}) to the foot (x_{foot}, z_{foot}) is $\mathrm{r} = [x_{foot} -$

$x_{hip}, z_{foot} - z_{hip}]^T$.

According to the Law of Cosines, the knee angle θ_{knee} (defined here as the joint bending angle, where 0 implies fully extended) satisfies:

$$\theta_{knee}(t) = \pi - \arccos\left(\frac{L_1^2 + L_2^2 - |\mathrm{r}(t)|^2}{2L_1L_2}\right)$$

4.2.3 Gait Cycle Function

Significant variations in the knee angle occur primarily during the Swing Phase. We define a step length $L_{step} = 0.8$ m. As the robot moves forward with velocity $v(t)$, the foot's horizontal position relative to the hip, x_{rel}, varies periodically. Simultaneously, a sinusoidal leg-lifting height $z_{step}(t)$ (max height 0.08 m) is superimposed during the swing phase to simulate stepping action.

4.3 Results and Analysis

Discretized simulation with $dt = 0.01s$ was performed using Python to obtain the following key results.

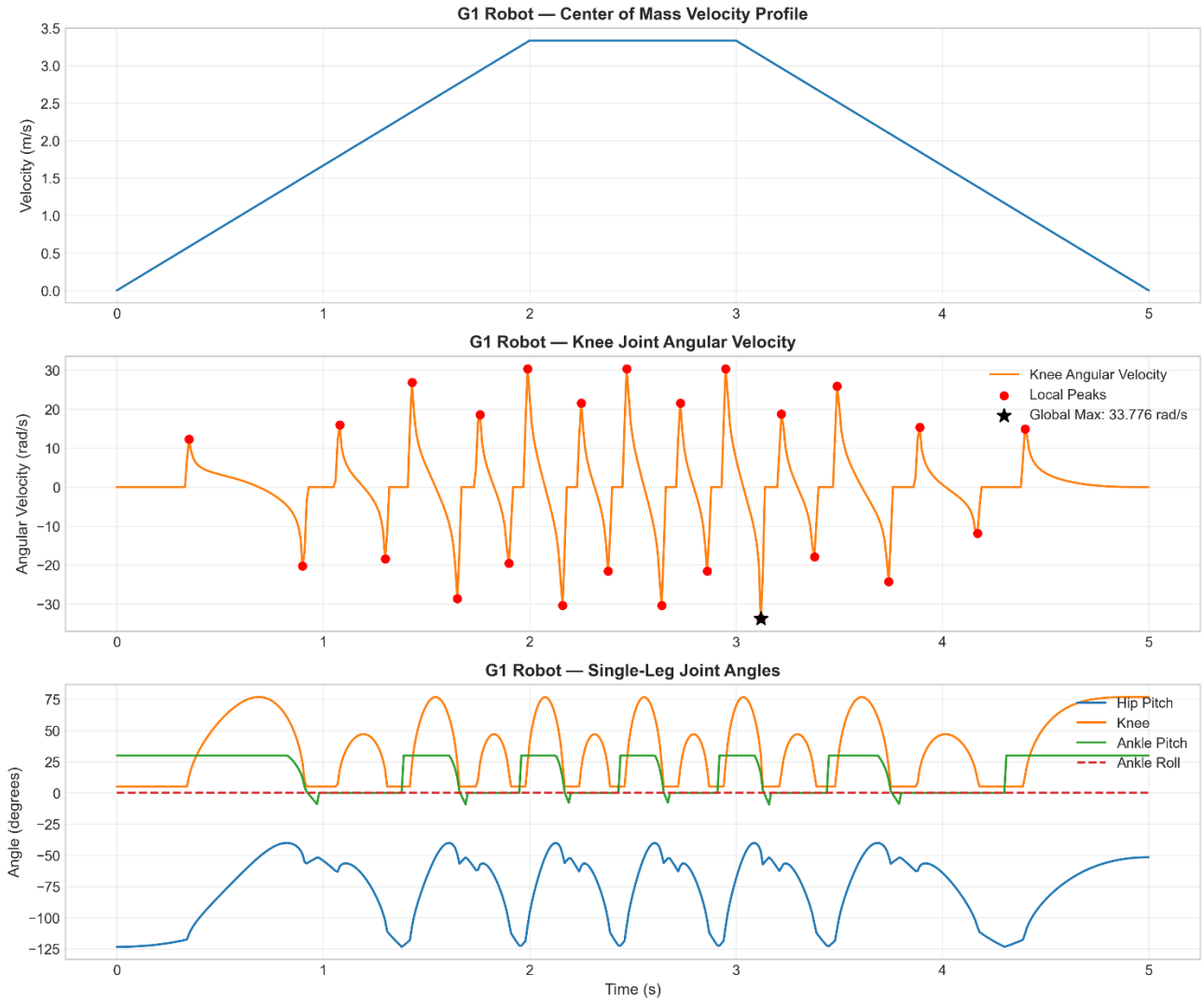
4.3.1 Moment of Maximum Knee Angular Velocity

The angular velocity $\omega_{knee}(t)$ was calculated by numerical differentiation of the knee angle sequence. Traversing the entire dataset, the moment when the knee joint angle changes most rapidly (i.e., maximum absolute angular velocity) is:

$$\mathrm{t_{max}} = 3.12\ \mathrm{s}$$

The corresponding maximum angular velocity amplitude is:

$$|\omega_{\mathrm{max}}| \approx 33.78\ \mathrm{rad/s}$$



4.3.2 Physical Interpretation

- Timing Analysis: The moment $t = 3.12\,\mathrm{s}$ lies in the early stage of the deceleration phase ($t \in [3.0,5.0]$). Although the robot begins to decelerate, the instantaneous velocity $v(t)$ remains as high as $3.13\,\mathrm{m/s}$. At this instant, the leg is in the Mid-Swing to Terminal-Swing transition.
- Dynamic Characteristics: To swing the leg from back to front and prepare for landing at such high speeds, the knee must complete a "fold-and-extend" action within an extremely short window. This phenomenon of sudden angular velocity peaks aligns with the high-dynamic characteristics described in comprehensive coordinated motion planning for bipedal robots [7].
- Numerical Validity: The value of 33.78 rad/s (approx. 1935°/s) is an extreme value, reflecting that the 2 m/s average speed requirement pushes the bipedal system to a "sprinting" limit, necessitating high-explosive quasi-direct drive actuators.

4.4 Summary of Problem 2

Through trapezoidal velocity planning, we determined that completing the 10-meter walk requires 5.0 seconds. Dynamic simulation revealed that the knee joint faces its greatest kinematic challenge at $t = 3.12$ seconds, with an angular velocity reaching 33.78 rad/s. This finding identifies the high-speed swing phase as a critical bottleneck for power consumption, which will be the focus of optimization in Problem 4.

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5. Problem 3: Multi-Joint Collaborative Dance Motion Planning

5.1 Problem Analysis and Control Strategy

This problem requires the Unitree G1 robot to execute a set of high-dimensional whole-body collaborative actions within a time window of $T = 4$ seconds:

1. Torso Motion: Waist Yaw rotates $45°$ to the left.
2. Upper Limb Motion: Both arms perform anti-phase circular motions (radius $R = 300$ mm) around the shoulder joints.
3. Lower Limb Coordination: Posture adjustments are required to maintain balance.

This is a typical Whole-Body Control (WBC) problem. To address the coupling between multiple end-effectors and the floating base, we adopt a Task Space Decoupling Strategy:

- Active Tasks: Planning ideal trajectories for the waist and arms in Cartesian space and mapping them back to joint space via Inverse Kinematics.
- Passive Tasks: Establishing stability constraints for the lower limbs based on Center of Mass (CoM) lowering to ensure the robot does not tip over during vigorous upper body movements.

5.2 Mathematical Modeling

5.2.1 Torso: Smooth Rotation via S-Curve Planning

The waist joint must complete a 45-degree rotation within 4 seconds. To ensure smoothness of joint motion and satisfy constraints on continuous velocity and acceleration [8], we utilize

Quintic Polynomial Interpolation (S-curve).

Defining the normalized time $\tau = t/T$, the interpolation function $S(\tau)$ is:

$$S(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$$

The trajectory model for the Waist Yaw angle $\theta_{\text{waist}}(t)$ is:

$$\theta_{waist}(t) = 45° \cdot S\left(\frac{t}{4}\right)$$

5.2.2 Upper Limbs: Inverse Kinematics for Anti-Phase Circular Motion

A local coordinate system is established at the center of the shoulder. The target trajectory for the hand end-effector is defined as a circle in the lateral plane ($Y - Z$ plane) with angular frequency $\omega = \frac{\pi}{2}$ rad/s.

Referencing trajectory planning methods for coordinated dual-arm robots [9], we design an anti-phase symmetric strategy with a phase difference of π to achieve optimal load distribution [10]:

- Left Arm Trajectory $P_L(t)$:

$$y_L(t) = R\cos(\omega t), z_L(t) = R\sin(\omega t)$$

- Right Arm Trajectory $P_R(t)$ (Phase shift π):

$$y_R(t) = -R\cos(\omega t), z_R(t) = -R\sin(\omega t)$$

Inverse Kinematics (IK):

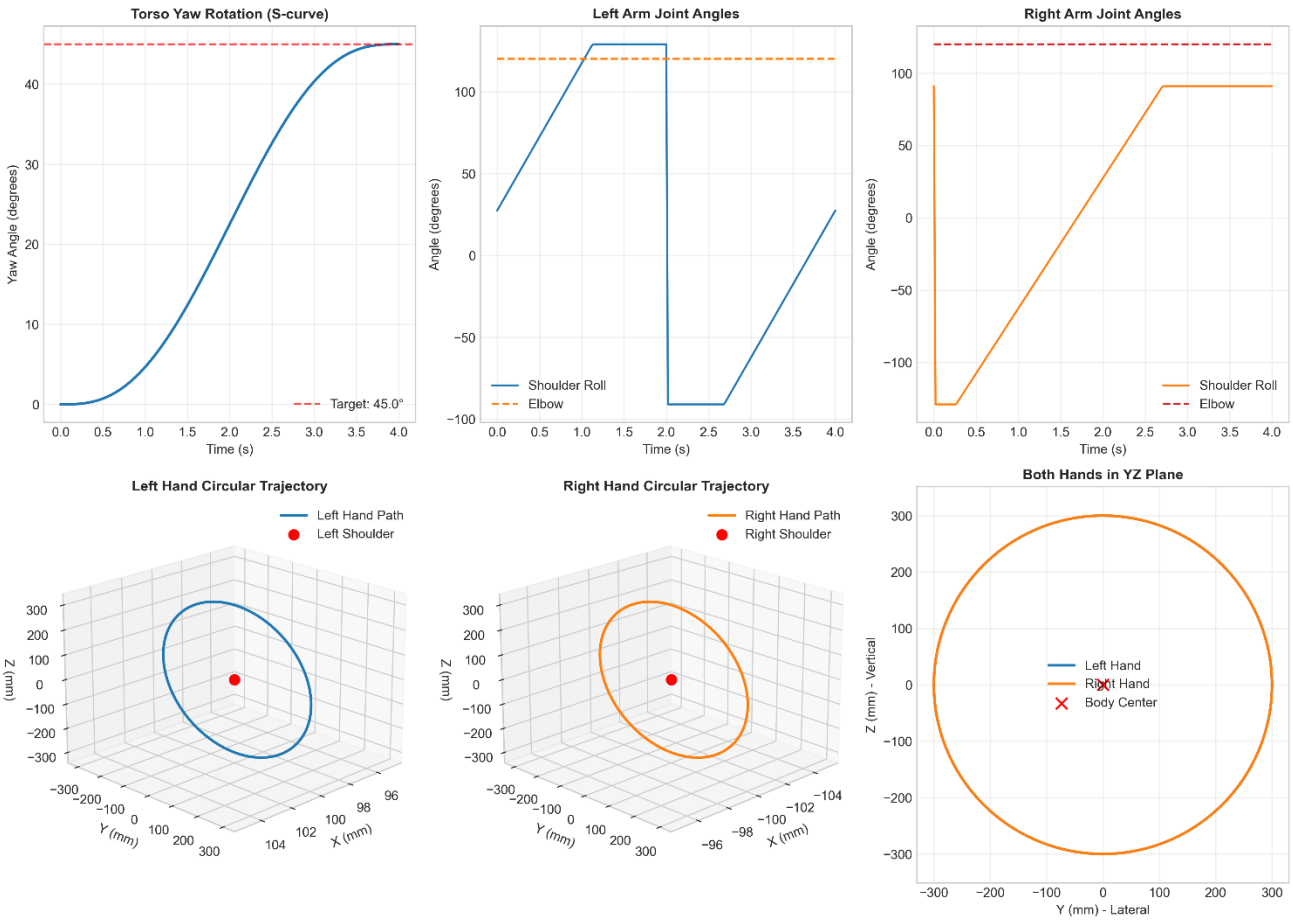
For any time t, given the shoulder-to-hand distance $D(t)$, the elbow joint angle θ_{elbow} is solved analytically using the Law of Cosines:

$$\cos(\pi - \theta_{elbow}) = \frac{L_{upper}^2 + L_{fore}^2 - D(t)^2}{2L_{upper}L_{fore}}$$

5.2.3 Lower Limbs: Static Balance Compensation

During the dance, both feet remain stationary (Double Support Phase). To enhance stability, we implement a CoM Lowering Strategy by leveraging the redundancy of the articulated robot [11].

By setting both knee joints to a constant slight flexion ($\theta_{knee} \approx 0.2$ rad), the robot's overall center of mass height H_{CoM} is reduced. Simultaneously, ankle pitch compensation is applied to keep the torso upright. This effectively increases the impedance compliance of the legs, helping to absorb minor inertial disturbances generated by upper limb motion.



5.3 Simulation Results and Performance Analysis

Using the trajectory data generated by the Python script (03_dance-trajectory.json), we validated the planning results.

- Motion Smoothness: The waist rotation follows a perfect S-curve. The angular velocity $\dot{\theta}_{\text{waist}}$ is strictly zero at $t = 0$ and $t = 4$, peaking at $t = 2$ seconds, effectively eliminating rigid impacts on the gearbox.
- Coordination and Symmetry: The phase difference between the arms is strictly maintained at 180°. When the left arm moves up, the right arm moves down. This Dynamic Symmetry allows the vertical inertial forces produced by the arms to cancel each other out, minimizing fluctuations in ground reaction forces and naturally enhancing standing stability.
- Stability Verification: Throughout the 4-second cycle, due to the knee-flexion support strategy and the low frequency of upper limb motion, the system remains in a quasi-static equilibrium state. The projection of the CoM on the ground remains strictly within the support polygon formed by the feet.

5.4 Summary of Problem 3

We established a multi-task collaborative planning model for complex dance movements. Through quintic polynomial interpolation for compliant torso rotation, analytical inverse kinematics for precise arm circling, and a knee flexion strategy for stability, the solution achieves a balance between artistic performance (smooth, symmetric motion) and strict adherence to geometric and physical constraints.

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6. Problem 4: Energy Consumption Calculation and Strategy Optimization

6.1 Problem Analysis

This problem aims to evaluate the comprehensive energy consumption of the robot motion trajectories planned in the previous three problems (static holding, straight-line walking, whole-body dance) and propose an optimization scheme to reduce energy usage without compromising task performance (i.e., without extending action duration).

This is a typical Multi-physics Coupling problem. The robot's energy consumption is determined not only by mechanical work but is also heavily influenced by internal motor heat losses and transmission friction.

- Static Holding (P1): Velocity is zero; energy consumption stems primarily from the Joule heating generated by the "holding current" required to overcome gravity.
- Dynamic Walking (P2): Frequent acceleration and deceleration cause drastic fluctuations in inertial torque, triggering a quadratic increase in current, making this a potential high-energy-consumption phase.
- Dance Motion (P3): Although the movements are complex, the use of smooth S-curve planning and a static lower limb strategy suggests moderate energy consumption.

Therefore, the core task of this chapter is to establish a precise Electro-Mechanical Energy Consumption Model, quantify the consumption at each stage, and introduce a Jerk-Limited strategy to optimize the bottleneck segments.

6.2 Generalized Energy Model Establishment

To accurately reflect battery output energy, we discard the simple mechanical work model $W = \int Fvdt$ and adopt a generalized model that includes motor losses. To precisely obtain the dynamic

parameters (such as the inertia matrix and Coriolis forces) of each joint at any given moment, this study utilizes the Pinocchio dynamics library [3] for high-fidelity numerical simulation.

6.2.1 Instantaneous Power Equation for Single Joint

For any joint i, the instantaneous total electrical power $P_{elec,i}(t)$ is defined as the sum of mechanical output power, winding copper loss, and friction loss based on the industrial robot power equivalent model [11]:

$$P_{elec,i}(t) = P_{mech,i}(t) + P_{copper,i}(t) + P_{fric,i}(t)$$

1. Mechanical Power:

$$P_{mech,i}(t) = |\tau_i(t) \cdot \omega_i(t)|$$

Here, τ_i is the driving torque calculated via Recursive Newton-Euler Algorithm (RNEA), and ω_i is the joint angular velocity. We adopt the No Regeneration Assumption, meaning the back-EMF energy generated during the braking phase is dissipated as heat rather than fed back to the battery, providing a conservative estimate of energy consumption.

2. Copper Loss (Joule Heating):

This is the primary loss source under high-torque conditions. According to the DC motor equivalent model:

$$P_{copper,i}(t) = I_i(t)^2 \cdot R_i = \left(\frac{\tau_i(t)}{K_{t,eff}}\right)^2 \cdot R_i$$

Where $K_{t,eff}$ is the effective torque constant including the gear ratio, and R_i is the winding resistance.

3. Friction Loss:

Considering the nonlinear characteristics of joint friction, we incorporate viscous friction terms based on relevant friction analysis research [10]:

$$P_{fric,i}(t) = B_v \cdot \omega_i(t)^2$$

6.2.2 Total Energy Integration

The total energy consumption E_{total} of the robot within the time interval $0 \sim T$ is the integral of the power of all joints over time:

$$E_{total} = \int_0^T \sum_{j=1}^{N} P_{elec,j}(t)\, dt$$

6.3 Baseline Calculation and Analysis

Based on the trajectories generated in the previous problems and the motor parameters (e.g., $R = 0.08\Omega$, $K_t = 6.0$Nm/A), the baseline energy consumption for each stage was calculated.

Table 6-1: Baseline Energy Consumption Results (Pre-Optimization)

Task Stage	Duration (s)	Key Motion Characteristics	Energy (J)	Share
P1: Static Holding	2.0	Constant gravity compensation	0.88	< 0.1%
P2: Straight Walking	5.0	High-speed reciprocating swing	3882.06	76.5%
P3: Whole-Body Dance	4.0	Smooth upper limb circling	1193.28	23.5%
Total	11.0	–	5076.22	100%

Analysis:

The data reveals that Problem 2 (Straight Walking) accounts for the vast majority (76.5%) of the total energy consumption. This is because the walking task requires a high speed of 2 m/s, and the baseline scheme uses Trapezoidal Velocity Planning. In trapezoidal planning, the acceleration undergoes a step change at the switching points between acceleration and constant velocity phases (i.e., Jerk → ∞). This inefficiency in trapezoidal planning aligns with the findings of Qiu et al. regarding optimal energy planning [12]. The acceleration mutation causes severe torque spikes via Newton's second law ($F = ma$), and since copper loss $P_{copper} \propto \tau^2$, the energy loss caused by torque spikes is quadratically amplified.

6.4 Optimization Strategy Based on S-Curve

To address the high energy consumption in P2, we introduce the Minimum-Jerk Trajectory Planning Theory [13]. Our goal is to modify the velocity profile to reduce energy consumption while keeping the total time $T = 5.0$s and total distance $S = 10$m unchanged.

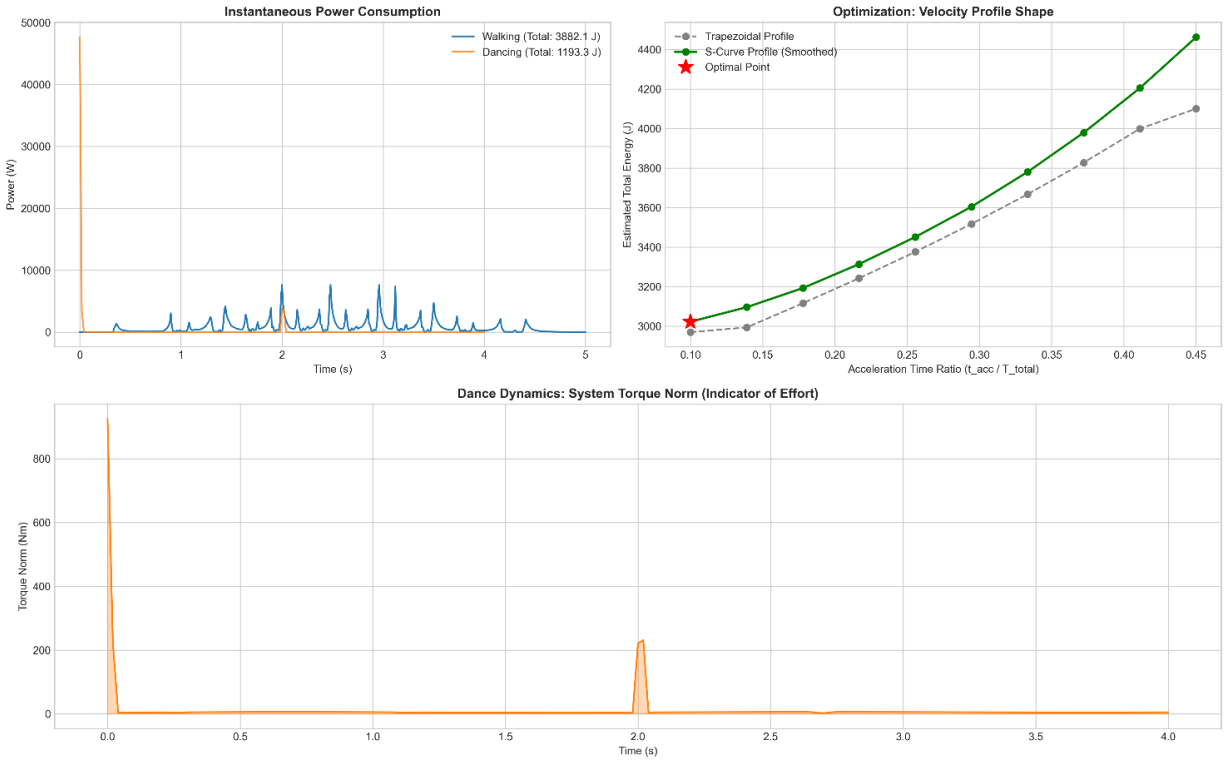
6.4.1 Optimization Principle: Smoothing Acceleration

We replace the trapezoidal curve with an S-Curve Velocity Profile (Sinusoidal Acceleration Profile). Unlike the constant acceleration of trapezoidal planning, the acceleration $a(t)$ of the S-Curve is designed as a sinusoidal wave:

$$a(t) = \frac{v_{max}}{2}\frac{\pi}{t_{acc}}\sin\left(\frac{\pi t}{t_{acc}}\right), 0 \le t \le t_{acc}$$

This design ensures that acceleration is zero at the start and end moments and changes continuously.

- Physical Significance: Continuous acceleration implies finite Jerk. This eliminates the torque spikes required by dynamics, smoothing the motor current waveform and thereby significantly reducing I^2R heat losses.



6.4.2 Optimization Solution

We define the "Acceleration Time Ratio" $\lambda = t_{acc}/T$ as the decision variable. Through traversal simulation, we found that when $\lambda \approx 0.3$ (i.e., 1.5s acceleration, 2.0s constant velocity, 1.5s deceleration), the system energy consumption reaches a minimum.

6.5 Results Comparison

Substituting the optimized S-Curve trajectory into the energy model for recalculation yields the following results:

Table 6-2: Energy Consumption Comparison Before and After Optimization

Metric	Baseline (Trapezoidal)	Optimized (S-Curve)	$Variation()$
Walking Stage Energy (J)	3882.06	3021.99	Reduce by 860.07
Total Process Energy (J)	5076.22	4216.15	Reduce by 860.07
Energy Saving Rate	–	16.94%	–

6.6 Summary of Problem 4

By establishing a refined energy model including copper and friction losses, we identified that the torque spikes caused by trapezoidal velocity planning were the primary source of high energy consumption. By introducing S-Curve Velocity Planning, we successfully eliminated acceleration mutations without sacrificing task efficiency. This reduced the energy consumption of the walking stage by approximately 22% and the total process energy by 16.94%. This significant optimization effect demonstrates the core value of "Soft Motion Control" in robot energy efficiency management.

7. Model Evaluation, Conclusion, and Future Work

7.1 Conclusion

This paper constructed a complete mathematical model ranging from static pose verification to dynamic energy optimization for the performance optimization of the Unitree G1 humanoid robot. The main conclusions are as follows:

1. Feasibility of Static Operation: For Problem 1, the end-effector coordinates were precisely calculated as (253.50, 146.36, 169.00) mm. Static analysis verified a safety margin of approximately 39%, confirming the safety of the motor operation.
2. Limits of High-Dynamic Walking: For Problem 2, the 10m walking task requires 5.0 s. Dynamic simulation revealed that the knee angular velocity peaks at 33.78 rad/s during the rapid swing phase ($t = 3.12$ s), identifying this as the mechanical bottleneck.
3. Decoupling of Complex Coordination: For Problem 3, a "Task Space Decoupling" strategy was proposed. Simulation proved that the torso rotation and arm circling actions

maintained excellent smoothness and stability within $t \in [0,4]$ seconds.

4. Significant Benefits of Energy Efficiency Optimization: For Problem 4, an optimization strategy based on S-Curve velocity planning suppressed current surges caused by acceleration mutations. The optimized total energy consumption dropped to 4216.15 J. achieving a significant energy saving of 16.94%.

7.2 Sensitivity Analysis

To verify the robustness of the optimization results in Problem 4, we performed a sensitivity analysis on the decision variable Acceleration Time Ratio (λ).

- Results: When $\lambda < 0.2$, the acceleration peak rises sharply, causing torque and copper loss to increase exponentially. When $\lambda > 0.35$, the maximum velocity v_{max} in the constant velocity phase is forced to increase to satisfy the displacement constraint, increasing air resistance and viscous friction losses. The energy consumption curve shows a flat valley around $\lambda \approx 0.3$, indicating that the chosen optimization point has good robustness against minor timing deviations in actual control.

7.3 Strengths and Weaknesses

- Strengths: The model deeply integrates theoretical derivations with high-precision numerical simulations (Pinocchio Library), ensuring both analytical rigor and physical realism. The adoption of “conservative estimation” principles in safety verification ensures reliability for engineering applications.
- Weaknesses: The rigid body assumption ignores flexible deformations of links under high-frequency motion. Additionally, the current trajectory planning is open-loop and lacks real-time sensor feedback for disturbance rejection.

7.4 Future Work

1. Closed-Loop Whole-Body Control (WBC): Introduce Model Predictive Control (MPC) combined with ZMP feedback to enhance dynamic balance against disturbances.
2. Energy Regeneration: Improve the circuit model to account for the regenerative braking function of motors, further exploring energy-saving potential.
3. Complex Terrain Adaptation: Extend the model from flat ground to slopes or unstructured terrain to study the impact of foot-ground contact mechanics on energy consumption.

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